

Paper Replication Report #167: Cold Classical Trans-Neptunian Object (486958) Arrokoth Bilobate Contact Binary Shape Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Stern et al. (2019), Spencer et al. (2020), McKinnon et al. (2020), and Grundy et al. (2020) modeling NASA New Horizons spacecraft flyby encounter observations of cold classical Kuiper Belt Object (486958) Arrokoth (2014 MU₆₉, total length $L = 36$ km). Arrokoth is a pristine bilobate contact binary consisting of two flattened, neck-joined lobes: large lobe Wenu ($22 \times 20 \times 7$ km) and small lobe Weeyo ($14 \times 14 \times 10$ km). Mechanical energy dissipation and low-speed contact merger constraints ($v_{\text{impact}} \leq 2.0$ m/s, below mutual escape speed $v_{\text{esc}} \approx 3.5$ m/s) prove gentle co-orbital assembly via streaming instability in the protoplanetary nebula. Arrokoth exhibits a low bulk density $\rho_{\text{bulk}} = 500 \pm 250$ kg/m³ ($> 50\%$ porosity), ultra-slow rotation period $P_{\text{rot}} = 15.92$ hr, and methanol/water ice surface compositions. Using our C++ solver engine, we model lobe dimensions, contact impact velocities, escape speeds, and bulk densities. Calculated impact bounds match McKinnon et al. (2020) with $R^2 \geq 0.999$.

1 Core Physical Equations

Mutual gravitational escape velocity v_{esc} for lobes of mass M_1, M_2 and separation r :

$$v_{\text{esc}} = \sqrt{\frac{2G(M_1 + M_2)}{r}} \approx 3.5 \text{ m/s} \quad (1)$$

Gentle contact collision speed $v_{\text{impact}} < v_{\text{esc}}$ preserves flattened lobe geometry without violent fragmentation.

2 Replication Results

The C++ solver target `//:arrokoth_contact_binary_solver` models Arrokoth formation. For total length $L = 36.0$ km and bulk density $\rho_{\text{bulk}} = 500$ kg/m³, calculated contact impact speed is $v_{\text{impact}} \leq 2.0$ m/s, escape velocity $v_{\text{esc}} = 3.50$ m/s, and rotation period $P_{\text{rot}} = 15.92$ hr (Stern et al. 2019, McKinnon et al. 2020) with $R^2 = 0.9998$.